Steven Varoumas

PhD in Computer Science

About me

Computer Science PhD (Sorbonne University) with expertise in embedded systems, compilers, and programming languages. Bilingual (French/English) and passionate about building efficient software solutions.

Areas of expertise

- Programming Languages
- Compilers and Virtual Machines
- Embedded Software

Spoken languages

- **French**: Fluent (mother tongue)
- **English**: Full proficiency
- **Spanish**: Limited

Tools & programming languages

OCaml

C & C++

git

Python

LLVM & MLIR

Tensorflow & XLA

Java

Haskell

stevenvar.github.io

github.com/stevenvar

Degrees

2019 **Ph.D. in Computer Science** · Sorbonne University (Paris, France)
"*High-level programming models for microcontrollers with scarce resources*"
- Lead Supervisor: T. Crolard (Conservatoire national des arts et métiers)
- Co-supervisors: E. Chailloux (LIP6 - Sorbonne University) and P. Trébuchet (ANSSI)

2015 **M.Sc in Computer Science** · Pierre and Marie Curie University (Paris, France)
Science and Technology of Software · mention *Bien* (with high honours)

2012 **B.Sc in Computer Science** · Pierre and Marie Curie University (Paris, France)
Programming and Software Development · mention *Assez Bien* (with honours)

Work experience

01/2021 - now **Compiler Engineer** · Huawei UK R&D
Cambridge, United Kingdom

- Developing advanced **MLIR transformations** and core compiler features to enable high-performance library optimisations.
- Extending **AI compiler** backends (TensorFlow/XLA) by implementing new optimisation passes and system features.

12/2019 – 09/2020 **Research Associate** · University of Kent
Canterbury, United Kingdom

- Developed an **automatic refactoring tool** for OCaml code and extension to new kinds of refactorings.
- Proposed and developed a method to perform automatised **equivalence checking** between two versions of a same program.

09/2019 – 11/2019 **Studies Engineer** · LIP6 laboratory · Sorbonne University
Paris, France

- Integrated my PhD research into the **LCHIP industrial project** in collaboration with the company *Clearsy*, which develops formal engineering methods and tools for **safety-critical systems**.
- Adapted a lightweight OCaml virtual machine for a secured runtime platform that uses PIC32 microcontrollers.

09/2018 – 08/2019 **PhD Student / Teaching Assistant** · Sorbonne University
Paris, France

- Co-creator and main developer of **a portable OCaml virtual machine** which can run on devices with very limited resources (**microcontrollers, FPGAs**,...).
- Created OCaLustre, **a synchronous programming language** for **concurrent embedded systems**.
- Developed a **static analysis tool** to compute worst-case execution time of an OCaLustre program, for critical embedded system.
- Taught approx. 400 hours of computer science classes for undergraduates.